

ONGOLE, India

WMO: 432210

Lat: 15.500N

Lon: 80.083E

Elev: 12

StdP: 101.18

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	21.3	12.7	9.2	28.1	14.6	10.4	28.4	2.5	28.7	2.4	28.6	0.6	N/A	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.1	41.2	25.8	39.4	26.0	37.9	26.0	28.6	34.9	28.1	34.0	27.7	33.3	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	31.3	26.7	22.3	31.1	26.4	21.9	30.9	92.7	35.0	90.6	34.3	88.8	33.3	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
3.3	2.8	2.4	DB	19.3	44.7	0.7	1.4	18.8	45.7	18.4	46.5	18.1	47.3	17.6	48.3	
			WB	15.2	29.8	1.2	0.7	14.3	30.3	13.6	30.7	12.9	31.1	12.0	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	25.9	27.3	29.9	31.9	34.3	33.3	31.5	30.9	30.4	29.9	27.7	26.2
	DBStd	3.10	0.90	1.05	1.16	1.22	2.42	2.50	2.20	1.97	1.70	1.84	1.41	1.10
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7283	494	484	616	658	754	699	665	649	613	617	529	504
	CDD18.3	4241	236	251	357	408	496	449	407	391	363	358	279	246
	CDH23.3	52119	1502	2076	4150	5597	7490	6745	5766	5298	4749	4416	2621	1709
	CDH26.7	27498	396	727	1856	3201	5027	4363	3327	2872	2402	2075	805	447
Wind	WSAvg	0.9	0.6	0.7	0.9	1.1	1.1	1.2	1.2	1.1	0.8	0.6	0.7	0.7
Precipitation	PrecAvg	1236	7	10	12	21	69	152	231	229	208	210	68	20
	PrecMax	1877	38	51	132	109	391	297	350	389	421	395	206	110
	PrecMin	897	0	0	0	2	12	60	138	127	68	24	1	0
	PrecStd	217	9	16	26	20	78	65	55	54	81	107	60	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	34.5	36.7	39.1	43.8	42.1	39.0	38.0	36.9	36.5	34.0	32.2
		MCWB	22.2	20.3	25.1	27.0	25.6	25.6	25.5	26.0	25.5	24.0	24.0	22.3
	2%	DB	30.5	32.6	35.1	37.2	42.1	40.4	37.6	36.8	35.7	35.2	32.6	30.9
		MCWB	22.2	22.4	24.7	26.9	25.6	25.7	25.2	25.4	25.4	24.5	24.0	22.9
	5%	DB	29.7	31.5	34.0	36.2	40.5	38.9	36.6	35.7	34.7	34.3	31.5	30.0
		MCWB	22.2	22.9	24.7	26.7	26.0	25.7	25.2	25.4	25.5	24.9	24.0	22.9
	10%	DB	29.0	30.5	33.1	35.2	38.5	37.5	35.4	34.6	33.7	33.2	30.4	29.2
		MCWB	22.1	22.8	24.6	26.6	26.5	25.6	25.1	25.4	25.7	25.2	24.1	22.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.4	27.2	28.6	29.3	28.8	27.4	27.6	27.9	27.6	26.5	25.3
		MCDB	28.6	30.4	33.1	35.0	35.6	35.0	31.9	32.5	32.5	31.8	29.8	28.6
	2%	WB	24.0	24.8	26.7	28.0	28.7	28.1	26.9	27.1	27.3	27.1	26.0	24.7
		MCDB	27.4	29.4	32.1	34.1	35.4	33.4	31.8	31.9	31.8	31.0	29.4	27.9
	5%	WB	23.5	24.4	26.2	27.7	28.4	27.6	26.5	26.7	27.0	26.7	25.7	24.2
		MCDB	27.1	28.9	31.2	33.3	34.7	33.0	31.7	31.6	31.3	30.7	28.9	27.5
	10%	WB	23.0	23.9	25.9	27.4	28.0	27.1	26.1	26.4	26.6	26.3	25.3	23.8
		MCDB	26.9	28.3	30.6	32.6	33.9	32.8	31.7	31.5	30.9	30.4	28.4	27.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	8.3	8.0	9.1	7.9	6.8	6.6	6.4	6.7	6.4	7.5
		MCDBR	8.5	9.2	9.0	8.7	11.4	9.3	8.1	7.8	7.7	8.2	7.7	8.2
		MCWBR	2.2	2.4	2.1	2.2	3.1	2.6	2.1	2.0	2.2	2.4	2.2	2.5
	5% WB	MCDBR	7.1	7.6	8.0	8.3	9.5	8.3	7.5	7.2	6.8	6.5	5.7	6.5
		MCWBR	2.2	2.3	2.1	2.2	2.9	3.0	2.4	2.5	2.4	2.4	2.1	2.3
Clear-Sky Solar Irradiance	taub		0.507	0.496	0.548	0.568	0.638	0.585	0.554	0.540	0.536	0.561	0.540	0.561
	taud		2.008	2.061	1.946	1.936	1.790	1.931	2.028	2.072	2.080	1.996	1.993	1.900
	Ebn at Noon		782	816	784	767	704	734	760	778	782	750	746	718
	Edh at Noon		170	168	193	195	223	191	174	168	166	177	171	185
All-Sky Solar Radiation	RadAvg		4.84	5.74	6.29	6.68	6.24	5.11	4.65	4.77	4.87	4.64	4.44	4.39
	RadStd		0.18	0.23	0.23	0.22	0.20	0.41	0.49	0.28	0.37	0.42	0.34	0.29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.28	+0.36	N/A	N/A	+0.27	+0.25	N/A	N/A	N/A	N/A

Nomenclature: See separate page